C PROGRAMMING

Lesson 12 — The return Keyword

Based on: Bro Code C Tutorial Series

1. What does return do?

In Lesson 11, functions used void — they performed a task but gave nothing back. The return keyword lets a function send a value back to whoever called it. Think of it like asking a question and getting an answer back.

```
// void function -- does something, gives nothing back
void greet() {
    printf("Hello!\n");
}

// return function -- does something AND gives a value back
int square(int num) {
    return num * num;   // sends the result back to the caller
}

int main() {
    int result = square(5);   // result = 25
    printf("%d\n", result);  // Output: 25
    return 0;
}
```

> **Key Idea:** return replaces the function call with the actual value. So square(5) becomes 25 wherever it is used in the code.

3. Return Type Must Match

The return type in the function definition must match the type of value being returned. Getting this wrong causes compiler warnings or errors.

```
// Returning int
int square(int num) {
    return num * num;       // int -- correct
}

// Returning double
double square(double num) {
    return num * num;       // double -- correct
}
```

```
// Returning bool
#include <stdbool.h>
bool isAdult(int age) {
    return age >= 18;        // bool -- correct
}
```

> **Rule:** int function must return int. double must return double. bool must return bool. Mixing types causes compiler warnings and unexpected behaviour.

4. return in main()

You have been writing return 0; at the end of main() since Lesson 1. Now you know exactly what it means.

```
int main() {
    // your program
    return 0;    // 0 = success, program finished correctly
}

// Non-zero return = something went wrong
int main() {
    return 1;    // signals an error to the operating system
}
```

5. Practice Problem

> **Practice: Temperature Converter**

Write a C program with two functions that convert temperatures, both using return.

Requirements:

- Write double celsiusToFahrenheit(double c) that returns the converted value
- Write double fahrenheitToCelsius(double f) that returns the converted value
- Formulas: $F = (C \times 9/5) + 32$ and $C = (F - 32) \times 5/9$
- In main(), ask the user for a temperature and which direction to convert, then print the result
- Also write a bool isBoiling(double celsius) that returns true if temp >= 100

Expected Output

```
Enter temperature: 100
Convert: 1) Celsius to Fahrenheit  2) Fahrenheit to Celsius
Choice: 1

100.00 C = 212.00 F
Is it boiling? Yes
```